Java Programming

CPT111 – Lecture 6

Brennan Jones

Xi'an Jiaotong-Liverpool University
西交利物浦大学

CPT111 Java Programming

Lecture 6

Introduction to Objects

Previously, in the last few weeks:

- Arrays and Loops
  - To store useful information in arrays
  - To store values in arrays - primitives and objects
  - To use loops to navigate through arrays

- Methods
  - Break code up into blocks
  - Using public static methods
  - Create smaller main methods

This Week:

- Objects
  - Representing the world as objects
- Constructor
  - Populating the instance variables
- Public and private
  - Should variables and methods be shared or kept private?
- Encapsulation
  - Data hiding
- Using objects within a array with for loops
- Static vs Non-static
- Comparing two objects

Learning Outcomes

- Relevant Learning Outcomes:

  A. Understand and appreciate the principles of using object-oriented programming techniques for the construction of professional robust, maintainable programs deployed to meet real world business goals;

  B. Design, write, compile, test, debug and execute object-oriented programs using an integrated development environment (IDE);

  C Effectively develop object-oriented programs as a member or a leader of a software development team with continuous development strategy supported by AI technology;

  D Implement object-oriented programming to represent, display, and manipulate data while mitigating security risks;

  E Evaluate legal, social, environmental, ethical, diversity, inclusion, and intellectual property issues related to object-oriented program development.

  F Apply knowledge of engineering management principles, commercial context, project and change management in object-oriented program development.

Object Oriented Programming

Representing things as objects

Primitives don’t have methods

- Primitive data types (**i**nt, **d**ouble, **b**oolean etc.) store an actual value
  - The variable holds that value

- Primitives have no methods
  - Only the value

Wrapper Classes

```
Integer.parseInt("3");
```

- Earlier, we introduced “Wrapper Classes”
- Useful for converting between String, Integers etc.
- A way of treating primitives as objects. We wrap the primitive to access useful methods

- Although an `int` is a primitive, an `Integer` allows us to treat it as an object, and so there are a number of methods, such as `Compare()`, `toString()`, etc. that we can access

Objects have methods

- Objects (**S**tring, **S**canner, **M**ath) are referred to by reference
  - The variable holds a memory address pointing to the object
- The object contains **properties** (variables) and **behaviour** (methods)

Objects

- Java is an object-oriented language
  - Nearly everything in Java is an object.
- The world is an object-oriented world
  - Nearly everything in the world is an object.

- Real-world objects have **states** and **behaviour**

Objects

- Objects in Java often represent objects in the real world (or e.g., a game)
- These objects have **state**: e.g., a character in a game could have
  - Location
  - Points
  - Strength
  - Weapons
  - *States can change*
- They would also have **behaviour**:
  - Walk
  - Fight
  - This ***behaviour*** *changes the* ***state***

Objects – key points

- Can store everything we need in objects
- Objects have **Attributes** - variables
- Objects have **Methods**
- Anything can be an object!

- Better to group connected data together as an object

Previously used Java Objects

String and Scanner

Existing Java Objects

- We have encountered several different objects so far. This includes `String`, `Math`, and `Scanner` objects
- The `String` object holds a piece of data – text – and has **methods** to **process** it.
- Similarly, the `Scanner` and `Math` objects also have data and methods

String Object

```
public class UsingStrings {

    public static void main(String[] args) {

        String someText;
        someText = new String();
        someText = "Hello Brennan";
        System.out.println(someText);
    }

}
```

- The variable **`someText`** is declared and holds an object of the **`String`** class.
- **`new String()`** creates – **instantiates** - a new **`String`** object
- This object is given a value of: `"Hello Brennan"`

String Methods

- `myString.length();`
- `myString.toLowerCase();`
- `myString.toUpperCase();`
- `myString.equals(anotherString);`
- `myString.equalsIgnoreCase(anotherString);`
- `myString.contains("java");`

Scanner Object - Review

- The `Scanner` object is a useful way to receive the `System.input` character stream, i.e. accessing characters that you type
  - Like a `String`, a `Scanner` is an object
  - It contains a number of useful methods
- Its purpose is to receive and handle keyboard input

Objects Summary

- Objects can have **methods**
- Can store **data**
- There are many **pre-written objects**
  - Code written by someone
  - Stored as a Java library
    - (Can import as needed)
  - They have methods that you can access
- You can make your own objects!

Data Objects

Creating your own objects

Data or Real-World Objects

- A very common type of object in Java is a data object
- Often represents a real-world object
  - such as a `Person`
    - It does not represent a whole Person
    - It holds the **pieces of data that we are interested in**.
- We can write a Java class to represent this object

Classes in Java

- Currently we have been working in a single class
  - Everything is static – a static class
    - **Static** means that a class is just a class
      - When you run it, there's one version of it
- However, we can use the same class to create **multiple objects**

The Person *class*

Defines ***Objects*** to hold simple data
about a person

Person Data Object

- If we want to store information about people, we can create `Person` objects
- Each `Person` has a number of different properties
  - Name
  - Sex
  - City
  - Date of birth
  - Id number
- We can store these in Java

A Person class

```java
package personproject;

public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;
```

So, how do we create all these different `Person` objects with their different data?

- The class defines the data that each `Person` object will hold
- There will be many `Person` objects (one for each person!)
- Each `Person` object will contain different data
- Each `Person` *object* is an **instance** of the `Person` *class*
- These variables are called **instance variables**

A Person class

```
public class Person {
```

- We start by declaring
- `public class _____`, and open with {
- Similar to the class files you have been creating so far
- Best to create it in a new Java file

Instance Variables

```
String name;
String gender;
String city;
String dateOfBirth;
int idNumber;
```

- Remember, each `Person` will be different, so they will store different values
- Note, these are not static!
- If they are static then no matter how many objects you create, they will overwrite
- Each `Person` object will contain different data.
- Each `Person` *object* is an **instance** of the Person *class*.
- These variables are called **instance variables**.

Our class so far

```java
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;
```

- No methods
- No main methods
- **How do we give each object values?**

Constructors

Making distinct objects

Constructor

- An object has special methods called “Constructors”
- Can be used to assign values when an object is created (“constructed”)
- Can also carry out any initial tasks
- It can have parameters
  - Whatever parameters you want!
  - Can be used to assign values to the instance variables

Creating a constructor

- In our `Person` class, we created a number of class variables
- But we did not give them values
  - As they are objects, each one we create can have different variable values
- It is the **constructor** that allows our initial objects to be different when they are first created!
- This is how we create **different people** with **the same class**

```
//constructor
public Person(String name, String gender,
              String city, String DoB, int id) {
```

Constructor

- When an object is created, it looks for a constructor
- A class can have more than one constructor
  - But it **should have at least one**
  - It will run this method **whenever a new object is created**
  - This is how we can pass variables into the object

A Constructor has no return type

The Constructor's name is **exactly the same as the class name**. So it is the only method to start with an uppercase letter.

```
//constructor
public Person(String name, String gender,
        String city, String DoB, int id) {
```

The Constructor should take one argument for each of the object's data fields. (It doesn't have to, but in our case it should.)

```java
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    //constructor
    public Person(String name, String gender,
            String city, String DoB, int id) {
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth =DoB;
        this.idNumber = id;
    }
```

- The Constructor has an argument for each object of data; for each instance variable.
- These are **local variables**, remember?
- So we assign the values to the actual **instance variables,** like this:

The `Person` class has no main method. That will be in a separate controller class:

```java
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    //constructor
    public Person(String name, String gender,
            String city, String DoB, int id) {
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth =DoB;
        this.idNumber = id;
    }
```

- The Constructor has an argument for each piece of data; for each instance variable.
- These are **local variables**, remember?
- So we assign the values to the actual **instance variables,** like this:

```java
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    //constructor
    public Person(String name, String gender,
            String city, String DoB, int id) {
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth =DoB;
        this.idNumber = id;
    }
```

- The Constructor has an argument for each piece of data; for each instance variable.
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```java
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    //constructor
    public Person(String name, String gender,
            String city, String DoB, int id) {
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth =DoB;
        this.idNumber = id;
    }
```

- The Constructor has an argument for each piece of data; for each instance variable.
- These are **local variables**, remember?
- So we assign the values to the actual **instance variables,** like this:

Our class so far

```
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    // Constructor
    public Person(String name, String gender,
            String city, String dOB, int id){
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth = dOB;
        this.idNumber = id;
    }

}
```

- Declare a class
- No main method
- Declare instance variables
- Create a constructor

this.objectInstanceVariable

Referring to an object variable

This Reference

- In our constructor, we have **2 different variables** with the **same name**
- One was created as an instance variable and can be seen **anywhere in the class**
- One is created **within the constructor function**, and contains the argument passed in, and only exists within the constructor
- This is confusing
- How can we solve this?

```
public Person(String name) {
    this.name = name;
}
```

This Reference

- One option: use different variable names!

```
public Person(String inputName) {

    name = inputName;
}
```

This Reference

- Another option: use `this.name` (use the ‘`this`’ keyword)
- When we use `this.`, we refer to the instance variable
  - So we can access this object’s instance/class variables
- `this.name` is the class variable
- `name` is the local variable;

```
public Person(String name) {

        this.name = name;
}
```

This Reference

Using `this.____` refers to **this particular object**

- It can refer to **its own variables and methods**
- Can call any of its own methods within the class
- Or any of its own variables
  - e.g., `int a = this.name;`
- Very useful in constructors!

```
this.name = “A call to the object”;
this.getName();
```

Our class so far

```
public class Person {
    String name;
    String gender;
    String city;
    String dateOfBirth;
    int idNumber;

    // Constructor
    public Person(String name, String gender,
            String city, String dOB, int id){
        this.name = name;
        this.gender = gender;
        this.city = city;
        this.dateOfBirth = dOB;
        this.idNumber = id;
    }

}
```

- Declare a class
- No main method
- Declare instance variables
- Create a constructor

Creating a Person Object

Creating from our Main class

Constructing a Person object

main() is NOT in the Person class.

```java
public class PersonProject {

    public static void main(String[] args) {

        Person p1;
        p1 = new Person("James Bond", "M",
                "London","1921-11-11", 007);

        Person p2 = new Person("Jackie Chan", "M",
                "Hong Kong","1954-4-7", 1);
```

In the Person class:

```java
//constructor
public Person(String name, String gender,
        String city, String DoB, int id) {
```

Creating our people

- We use our `Person` class to create objects
- In our `main` method, we create a new `Person` object
  - Just like when we created a new `Scanner` etc.
- We choose arguments that match the constructor

```
public static void main(String[] args) {

        Person p1 = new Person("James Bond", "M",
        "London", "1921-11-11", 007);

}
```

We now have two different classes

Our **main** class has the `main()` method.

- This is sometimes called the ***controller*** class.
- It is (usually) static.

Only one main class exists in the application. It has no **constructor**.

The **main** class can have or contain lots of objects – e.g., data objects such as `Person` objects.

- Each `Person` object holds data for one person.

Our `Person` class is not static.

- It defines `Person` data objects: their data fields and methods.
- It has a constructor used to construct `Person` data objects, each with ***different*** data.
- Many different `Person` data objects will exist.

So now our Person class defines a very basic data object
(actually, right now it is more of a structure, or struct – it has no methods)

Creating our people

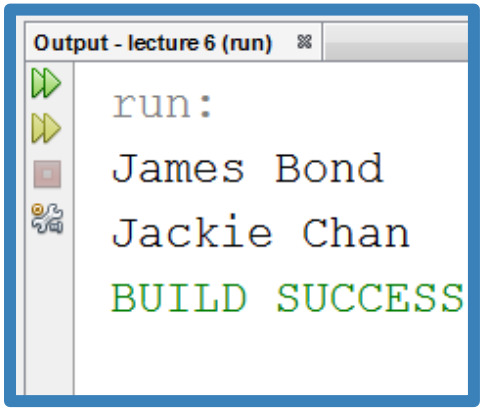
- Can use the same class to now create multiple people
- The same class can create different objects

```
public static void main(String[] args) {

        Person p1 = new Person("James Bond", "M",
        "London", "1921-11-11", 007);

        Person p2 = new Person("Jackie Chan", "M",
        "Hong  Kong", "1954-4-7", 1);
}
```

Creating our people

- Can access our data using the data fields in `Person`
- Note that the `Person` code is unchanged
- But by storing things as an object, we can have **multiple objects** using the **same class**

```
System.out.println(p1.name);
System.out.println(p2.name);
```

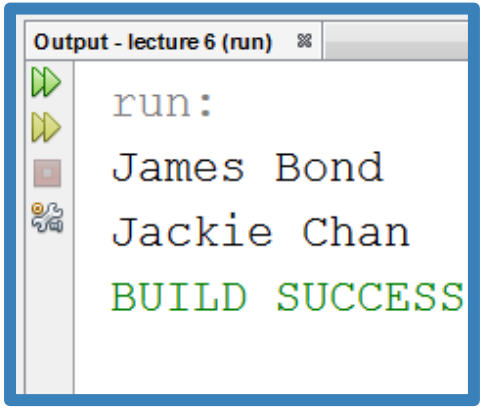


Creating our people

- Can access our data using the data fields in `Person`
- Note that the `Person` code is unchanged
- But by storing things as an object, we can have **multiple objects** using the **same class**

```
System.out.println(p1.name);
System.out.println(p2.name);
```

Right now, anyone can access the data fields directly. However, anyone can also change them. We do not want that!



Public and Private

Data hiding

Public and Private

- So far, we haven't really cared
- Public or private, both work within a class
  - can create `public void`, or `private void` methods
- With multiple classes, there IS a difference
- If it is PRIVATE, can only be accessed by that class
  - Useful for **internal methods**
  - something that only needs to be done within the objects
- If it is PUBLIC, then it can be accessed by other classes
  - Useful for **object methods that need to be accessed**

Public and Private

- As a good habit, things should be private by default for data security
- variables, methods, should be private, unless we need to access it from a separate class
- We need to access our constructor and any other methods that we need
  - They should therefore be public!

Encapsulation

- **All attributes should be private**
- Internal workings should be private
  - The object should only share what it needs
- Can use getter methods to access some private attributes

```
private String name;
```

```
public String getName() {
        return name;
    }
```

```java
public class Person {
    private String name;
    private String gender;
    private String city;
    private String dateOfBirth;
    private int idNumber;
```

Make the fields private

Private instance variables and private methods can only be accessed from **within the same class**.

- OK, great… but now nobody can read the data in our data object!

```java
public String getName() {
    return name;
}

public String getGender() {
    return gender;
}

public String getCity() {
    return city;
}

public String getDateOfBirth() {
    return dateOfBirth;
}

public int getIdNumber() {
    return idNumber;
}
```

- Add accessor methods, or ‘getters’
- Public methods to return private data.
- **Hiding the data and other internal workings** of the class and objects, and **only accessing them through public methods**, is called **encapsulation**.
- Encapsulation is one of the core principles of OOP, and Java.

We use the getter methods to get the data:

```
public class PersonProject {

    public static void main(String[] args) {
        Person p1;
        p1 = new Person("James Bond", "M",
                "London","1921-11-11", 007);
        Person p2 = new Person("Jackie Chan", "M",
                "Hong Kong","1954-4-7", 1);

        System.out.println("Name = "+p1.getName());
```

```
run:
Name = James Bond
```

So, back in our main class, we can access the data fields in our data objects using the 'getter' methods. **But we can't change the data.**

We use the getter methods to get the data:

```
public static void main(String[] args) {

        Person p1 = new Person("James Bond", "M",
        "London", "1921-11-11", 007);

        Person p2 = new Person("Jackie Chan", "M",
        "Hong Kong", "1954-4-7", 1);

    System.out.println(p1.getName());
    System.out.println(p2.getName());
}
```

We can use the ‘getter’ methods to access our variables from the main class.

A little history...

Previously: we would also have 'setter'; and these are still common.

- However, the trend today is to have **immutable** objects: objects that cannot be changed after they are created.
- If you want to change any data, you have to make a new object

```java
public void setName(String name) {
    this.name = name;
}

public void setGender(String gender) {
    this.gender = gender;
}

public void setCity(String city) {
    this.city = city;
}
```

Why? Parallel or concurrent processing is much more common today, and immutable objects are safer for this

Can add **checking** for valid values ***before*** setting the value

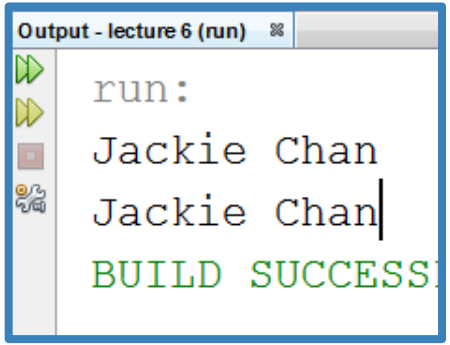
Static

- What about the `static` keyword?
- `static` means that there is **only one instance** for the entire class
  - So, even if we create 100 objects, if we use a `static` variable for person name, then **each person will have the same name**
- If it is `static`, each new object will overwrite the previous variable
- Can be useful

Static

```
public static void main(String[] args) {

    Person p1 = new Person("James Bond", "M", "London",
                           "1921-11-11", 007);

    Person p2 = new Person("Jackie Chan", "M", "Hong Kong",
                           "1954-4-7", 1);

    System.out.println(p1.getName());
    System.out.println(p2.getName());
}
```

```
public class Person {

    private static String name;
    private String gender;
    private String city;
    private String dateOfBirth;
    private int idNumber;
```

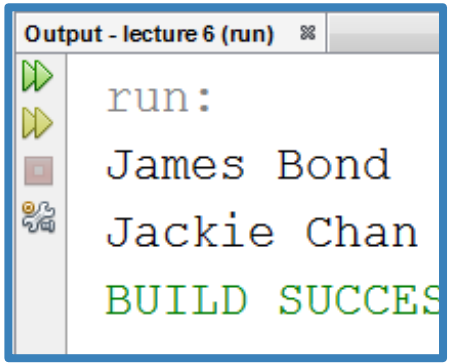


Static, so second person declaration overwrites first, meaning BOTH are Jackie Chan

Non-Static

```
public static void main(String[] args) {

    Person p1 = new Person("James Bond", "M", "London",
                           "1921-11-11", 007);

    Person p2 = new Person("Jackie Chan", "M", "Hong Kong",
                           "1954-4-7", 1);

    System.out.println(p1.getName());
    System.out.println(p2.getName());
}
```

```
public class Person {

    private String name;
    private String gender;
    private String city;
    private String dateOfBirth;
    private int idNumber;
```



Not static, so both names are stored as separate instance variables, and are different

Use of Static

- Our main method is `static`, as **there is only one main method** in the entire program
- In previous weeks, our methods were `static`, as we were working with a single class, and not objects
- Our previous variables were `static` for the same reason
- But with our People, we are working with objects, so we do not use `static`

Use of Static – Good use

Where might we use `static`?

- If we want to count how many objects created
  - How many `Person` objects?
- Can create in `main` method
  - OR can use a `static` variable to count each…

What's next?

- We now have a functioning data class
  - Private fields (instance variables)
  - Public getters to access them
  - A constructor to create new data objects
- It can store all data about a person
  - A way of grouping information about 1 person together
  - Better than lots of variables, or even arrays!
- We will add more features, for a really good data object

Primitive variables and Object variables

== and .equals()

Variables pass a *value* or a *reference*

- `int` and `double` are variables holding **primitive** data
- A `String` is a variable pointing to an **object** (String)
- **What is the difference?**
  - Primitive variables *hold a value*
  - Object variables *hold a reference, a pointer to the memory address of the object*

```
int myNumber = 7;
int newNumber = myNumber;
myNumber ++;
```

Here, the **value** of `myNumber` (7) is assigned to `newNumber`. The two variables are completely unrelated and unconnected

Objects vs primitives

```
Person[] people = new Person[2];
people[0] = p1;
people[1] = p2;
System.out.println("Array is " + people);
```

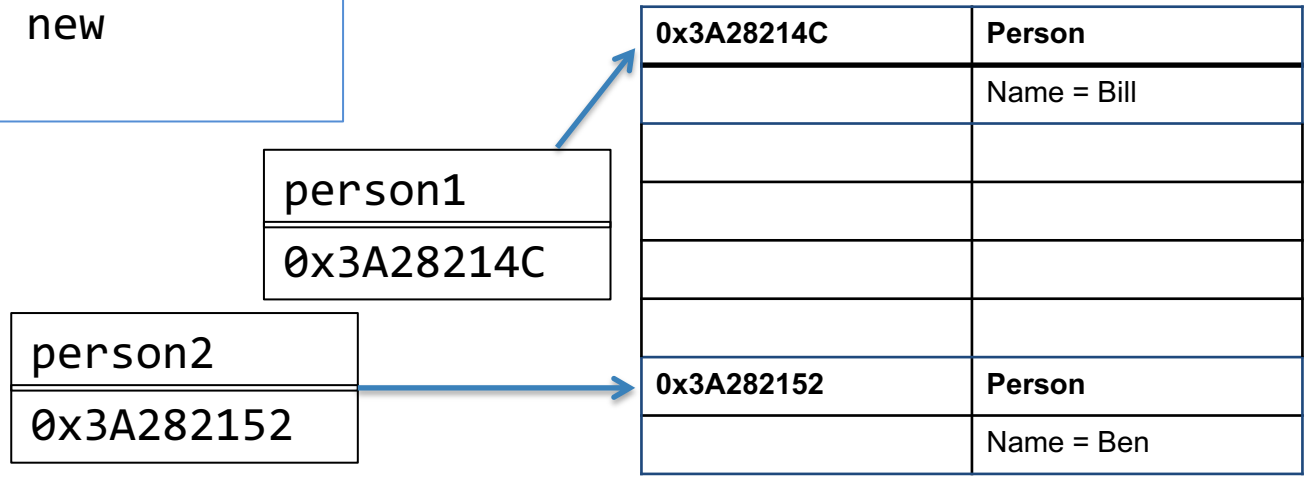
An array is **not a primitive**

- What happens when you try to print out whole array without looping through it?
  - It outputs the reference

```
Array is
[Lecture6.Person;@15db9742
```

Object variables: pointers

```
Person person1 = new
Person("Bill");
Person person2 = new
Person("Ben");
```



Object variables hold the **address in memory of the object**.

- They are **completely separate** from the object, and do not know or hold any of the object's values.
- They '**point**' to an address in memory – in some languages they are called pointers.

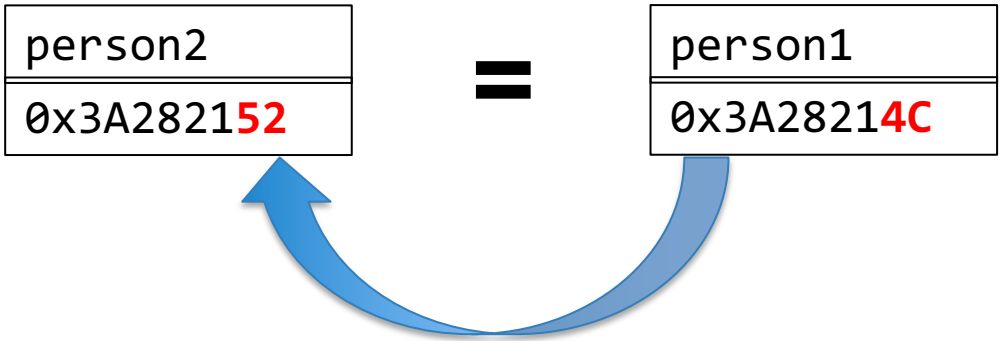
Pass by reference

person1
0x3A28214C

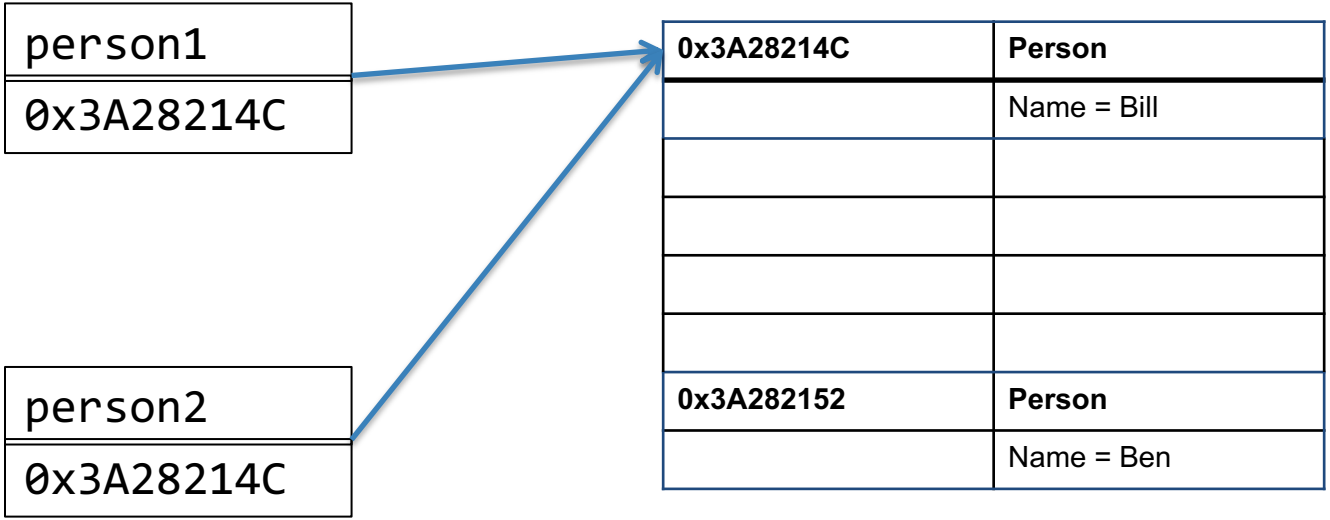
person2
0x3A282152

```
person2 = person1;
```

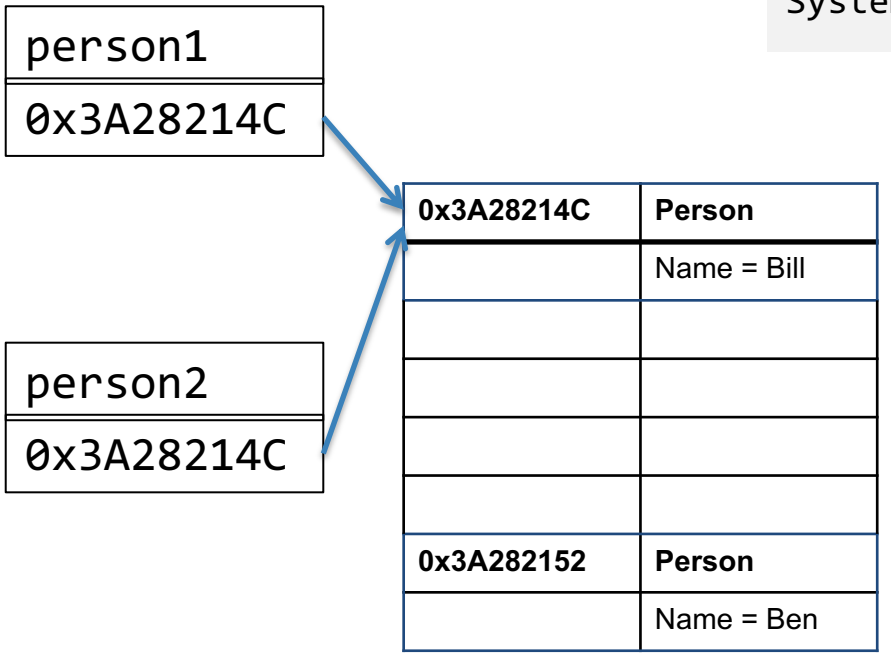
The memory address – **reference** – is copied from person1 to person2. **No *values* are copied at all.**



Now person1 and person2 point to the same Person object



```
Person person1 = new Person("Bill");
Person person2 = new Person("Ben");

person2 = person1;

person1.setName("David");

System.out.println("Name is " + person2.getName());
```



What is the output of this code?

```
Name is David
```

== and .equals()

double number1
3.214

==

double number2
3.214

True

Primitive variables hold a value. == compares the values, and returns TRUE if they are the same.

```
Person p1 = new
Person("John");
Person p2 = p1;
```

p1
0x00000001

==

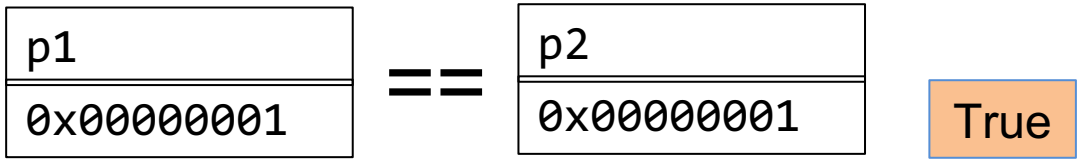
p2
0x00000001

True

Object variables hold an address. == compares the address, and returns TRUE if they are the same
(You could say that it is comparing the value of the address)

== and .equals()

```
Person p1 = new
Person("John");
Person p2 = p1;
```



- What if they hold the same values, but different addresses?
  - i.e. all the values are the same, but are stored as different variables?

```
Person p1 = new Person("James Bond", "M", "London", "1921-11-11", 007);

Person p2 = new Person("James Bond", "M", "London", "1921-11-11", 007);


System.out.println(p1 == p2);
```

How do we compare our Person objects?

.equals()

- In Java, objects have an `.equals()` method.
  - E.g. Strings: `variable.equals("variable")`
- In Java, we can use inheritance to "override" methods, and we can create an `equals` method
  - Inheritance will be discussed next week
  - `.equals()` is more advanced Java and will **not** be covered in this module (covered in ***CPT204 Advanced OOP***)

Same Person

- We can make an assumption that two people are the same if their `name`, `id`, and `gender` are the same
- We can therefore create a method
  - Return a `boolean` (`true` if same, `false` if different)
  - Use the getters to compare
  - Compare '`this`' object

```
public boolean samePerson(Person that) {
        return true;
}
```

Same Person

- Two people, ‘this’ and ‘that’
- We can compare ‘this’ person, i.e., the object, to ‘that’ person, i.e. the `Person` passed in
- Use the getters

```
Person p1 = new Person("James Bond", "M", "London", "1921-11-11", 007);
Person p2 = new Person("Jackie Chan", "M", "Hong Kong", "1954-4-7", 1);
boolean sameP = p1.samePerson(p2);
```

```
if(!this.name.equals(that.getName())) {
    return false;
}
```

Same Person

```
Person p1 = new Person("James Bond", "M", "London",
"1921-11-11", 007);
Person p2 = new Person("Jackie Chan", "M", "Hong Kong",
"1954-4-7", 1);
boolean sameP = p1.samePerson(p2);
```

Same Person

- `this.name = "James Bond”;`
  `that.getName() = "Jackie Chan”;`
- If they are not equal, return `false`

```
if(!this.name.equals(that.getName())){
            return false;
}
```

Same Person

- If any attributes are not matching return `false`
- If all attributes match, return `true`

```
public boolean samePerson(Person that){
    if(!this.name.equals(that.getName())){
        return false;
    }
    if(!this.gender.equals(that.getGender())){
        return false;
    }
    if(this.idNumber != that.getIdNumber()){
        return false;
    }
    return true;
}
```

The .toString() method

Displaying information

Anything else?

- Remember that objects are **pointers**, not values
- All Java objects also have a `toString()` method.
  - By default, it is not very useful.

```
System.out.println("Person 1 " + p1 );
```

```
Person 1
lecture5.Person@15db9742
```

Anything else?

- We need to define a `toString` method in our data class.
- It should print the values of all the data fields – instance variables – in the object

```
public String toString() {
    return "Person: name=" + name
            + ", gender=" + gender
            + ", city=" + city
            + ", idNumber=" + idNumber;
}
```

```java
public class PersonProject {

    public static void main(String[] args) {
        Person p1;
        p1 = new Person("James Bond", "M",
                "London","1921-11-11", 007);
        Person p2 = new Person("Jackie Chan", "M",
                        "Hong Kong","1954-4-7", 1);
        System.out.println("p1 = "+p1.toString());
        System.out.println("p2 = "+p2);
```

```
run:
p1 = Person: name=James Bond, gender=M, city=London, idNumber=7
p2 = Person: name=Jackie Chan, gender=M, city=Hong Kong, idNumber=1
```

Looking good!

Now we have a `Person` data class. It has:

- Private data fields – instance variables
- Constructor
- Public getter methods
- It is encapsulated
- It is immutable
- `toString()` method to print the object's values

Combining with Arrays

- Create an array of 3 people:

```
Person[] people = new Person[3];
```

- Instantiate our array of people
- Can use our objects we created earlier, or create new ones

```
people[0] = p1;
people[1] = p2;
people[2] = new Person("Diana Rigg", "F", "London", "1943-8-21",2);
```

For loops

- Now we can navigate through our array similar to before
- For example, if we want to show everyone's name

```
for (int i = 0; i < people.length; i++) {
    System.out.println("Name is " + people[i].getName());
}
```

```
Output - Lecture5 (run)
run:
Name is James Bond
Name is Jackie Chan
Name is Diana Rigg
BUILD SUCCESSFUL (total time: 0 seconds)
```

For loops

- Can use `toString()` method to display all

```
for (int i = 0; i < people.length; i++) {
    System.out.println("Name is " + people[i].toString());
}
```

```
utput - Lecture5 (run)
run:
Name is Person: name= James Bond , gender= M , city= London , date of birt
Name is Person: name= Jackie Chan , gender= M , city= Hong Kong , date of
Name is Person: name= Diana Rigg , gender= F , city= London , date of birt
BUILD SUCCESSFUL (total time: 0 seconds)
```

Thank you for your attention !

- In this lecture, you have learned about:
  - Objects, Attributes and Methods
  - Constructor and `this`.
  - Public vs Private
  - Static vs Non-static
  - Comparing two objects
  - Objects and Arrays